

NO. OF FUNCTIONAL FEATURES INCLUDED IN THE SOLUTION

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
Team Members	G Uday Kumar, Kavali Anu, Pavan Kumar Sowdepalli, Dasarayyagari Sravan Kumar
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

This document lists all the functional features included in the project solution[cite: 20]. Each feature is described with its purpose, implementation status, and contribution to the overall solution[cite: 20].

Functional Features Overview

Metric	Count / Value
Total Features Planned	8[cite: 20]
Total Features Implemented	6[cite: 20]
Core / Must-Have Features	5[cite: 20]
Additional / Nice-to-Have Features	2[cite: 20]
Features Tested & Verified	6[cite: 20]

Detailed Functional Features Matrix

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
1	User Registration	Allows new farmers, researchers, and stakeholders to create an account with personal details and store it in the database.[cite: 20]	Authentication Module[cite: 20]	Done[cite: 20]	0.50[cite: 20]
2	User Login	Allows registered users to securely login using email and password credentials.[cite: 20]	Authentication Module[cite: 20]	Done[cite: 20]	0.50[cite: 20]
3	Soil & Climate Input	Collects detailed environmental parameters (N, P, K, temperature, humidity, pH, rainfall) through a structured field form.[cite: 20]	Frontend Module[cite: 20]	Done[cite: 20]	0.75[cite: 20]

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
4	Data Preprocessing	Cleans, handles missing parameters, encodes and scales the biochemical vectors before feeding them to the machine learning core.[cite: 20]	Preprocessing Module[cite: 20]	Done[cite: 20]	0.75[cite: 20]
5	Crop Optimization Core	Predicts and suggests the most suitable high-yield crop match using multi-class trained machine learning classifiers.[cite: 20]	ML Prediction Module[cite: 20]	Done[cite: 20]	1.00[cite: 20]
6	Recommendation Display	Displays the generated crop compatibility index, confidence score, and productivity analysis summaries layout.[cite: 20]	Frontend Module[cite: 20]	Done[cite: 20]	0.75[cite: 20]
7	Historical Logs Log	Stores and displays past evaluation metrics and crop suggestion history logs for the active user.[cite: 20]	Database Module[cite: 20]	In Progress[cite: 20]	0.50[cite: 20]
8	Stakeholder Dashboard	Allows researchers and administrators to view system usage distribution, regional metrics, and overall summary stats.[cite: 20]	Admin Module[cite: 20]	In Progress[cite: 20]	0.25[cite: 20]

Feature Category Breakdown

S.NO	Category	Features in Category	Example Features
1	User Interface (UI)[cite: 20]	3[cite: 20]	Login Page, Farm Parameters Wizard Form, Result Visualizer View[cite: 20]
2	Backend / Application Logic[cite: 20]	3[cite: 20]	Soil Data Preprocessing, Crop Matching Engine, Business Rules Processing[cite: 20]
3	Database / Persistence Storage[cite: 20]	2[cite: 20]	Farmer Profile Storage, Longitudinal Query History Records[cite: 20]
4	API / Intelligence Integration[cite: 20]	1[cite: 20]	Machine Learning Core Inference Pipeline Integration[cite: 20]
5	Security / Authentication[cite: 20]	2[cite: 20]	User Registration, Secure Route Gateways[cite: 20]